STRATHMORE BUSINES SCHOOL
MASTER IN BUSINESS ADMINISTRATION
MAKE-UP CAT
MBA 8104 QUANTITATIVE TECHNIQUES FOR BUSINESS

DATE: Friday 17th July 2026 **TIME:** 2 Hours

INSTRUCTIONS:

1. This paper consists of 4 questions. Answer **Question ONE** and any other **TWO** Questions

QUESTION ONE (10 MARKS)

(a) An issue that faces individuals investing for retirement is allocating assets among different investment choices. Suppose a study conducted 10 years ago showed that 65% of investors preferred stocks to real estate as an investment. In a recent random sample of 900 investors, 300 preferred real estate to stocks. Are these new data sufficient to allow you to conclude that the proportion of investors preferring stocks to real estate has declined from 10yeard ago? Conduct your analysis at the $\alpha = 0.02$ level of significance. (4 Marks)

(b)A Production manager of a manufacturing firm believes that items produced by process B on average last longer than items produced by process A as measured by a standardized test. To test this belief, random samples of finished items produced by the two processes were tested and yielded the results in the output below.

Descriptive Statistics:

Variable	Total Count	Mean	StDev
Pop. A	40	52.30	9.23
pop. B	37	69.47	11.78

The manager believes that the relevant populations of scores are normally distributed, with equal, although unknown variances.

(i) Write down the null and alternative hypotheses (1 Mark)

(ii) State type I and type II error for this problem (2 Marks)

*(iii) At the 0.01 level of significance test whether her belief is valid (3 Marks)

QUESTION TWO (10 MARKS)

(a) Items coming from a production line are categorized as good (G), slightly blemished (B), and defective (D), and the percentages are 80%, 15% for good and for slightly blemished, respectively. Suppose that two items will be selected randomly for inspection and the selections are independent.

(i) Find the probability that at least one of the items is slightly blemished. (2 Marks)

(ii) Find the probability that neither of the items is good (2 Marks)

(iii)Suppose that one of the selected items is good, what is the (conditional) probability that both of them are good? (1 Mark)

(b) In a factory there are two machines manufacturing bolts. The first machine manufactures 75% of the bolts and the second machine manufactures the remaining 25%. From the first machine 5% of the bolts are defective and from the second machine 8% of the bolts are defective. A bolt is selected at random, what is the probability the bolt came from the first machine, given that it is defective? **(5 Marks)**

QUESTION THREE **(10 MARKS)**

A Tech company that services microcomputers is interested in developing a regression model that will assist them in manpower planning. In particular, they want a model that describes the relationship between the time a service person spends on a preventive maintenance service call to a customer, y, and two independent variables: the number of microcomputers to be service, X_1, and the service person's number of months of experience in preventive maintenance, X_2. Company records were sampled and the summary data below were obtained.

Regression Analysis:

Predictor	Coef	SE Coef	T	P
Constant	2.405	1.248	1.93	0.095
number of microcomputers	1.4261	0.1406	10.14	0.000
experience (Months)	-0.3663	0.1317	-2.78	0.027

S = 1.51504   R-Sq = 94.0%   R-Sq(adj) = 92.3%

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	251.60	125.80	54.81	0.000
Residual Error	7	16.07	2.30		
Total	9	267.67			

(i) Write the regression model **(1 Mark)**
(ii) Interpret the coefficients **(3 Marks)**
(iii) Interpret the value of coefficient of determination R-square above **(2 Marks)**
(iv) From the above output, what is the sample size in this study? **(1 Mark)**
(v) Write the hypothesis and deduce on the test for goodness of fit **(3 Marks)**

QUESTION FOUR **(10 MARKS)**

(a) A telecommunications company is planning to market a new product. According to the analysis made by the financial department of the company, it will earn an annual profit of $ 6.5 million if this product has high sales, annual profit of $ 2.2 million if the sales are mediocre and it will lose $ 1.5 million a year if the sales are low. The probabilities of these three scenarios are 0.30, 0.50 and 0.20 respectively. Let X be the profits (in millions) earned per annum by the company from this product. Calculate the mean and standard deviation **(4 Marks)**

(b) Analysis of the rate of turnover of employees by a personnel manager produced the following table showing the length of stay of 200 employees who left the company for other employment

Grade	Length of employment (years)			
	0-2	2-5	More than 5	
Managerial	4	11	6	21
Skilled	32	28	21	81

Unskilled	25	23	50	98

Using 0.01 level of significance test whether there is sufficient evidence to believe that there is a relationship between grade and length of stay in employment. **(6 Marks)**